Supplementary material: Incremental value of failure forecasts beyond a health-index landmark in conservative turbofan removal on C-MAPSS

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This supplement specifies the development comparators and empirical-support diagnostics used in the turbofan removal comparison.

S1 Health-index development comparators

The development identifiers P03–P06 refer to archived exploratory pilots: minimum departures (P03), initial tolerance (P04), empirical and terminal-reconstruction comparisons (P05), and transfer/refitting in FD003 (P06). These pilots precede the main-protocol freeze.

For sensor j, the causal summary is $\bar{x}_{ij,15}(t) = 15^{-1}\sum_{u=t-14}^{t} x_{ij}(u)$. Orientations, population scales and fitted coefficients are learned outside the evaluation engines; each engine supplies its own initial sensor reference. P03 investigates sensitivity of the detection assumptions while holding prognostic predictions fixed, together with an initial HI proxy. P04 introduces a tolerance for initial sensor variation. For oriented departure $a_{ij}(t) = d_j[\bar{x}_{ij,15}(t) - b_{ij}]$, with $b_{ij} = \bar{x}_{ij,15}(20)$ and $d_j \in \{-1, 1\}$ learned from fitting engines, the margin-based candidates take the form

$$\widehat{H}_i^{\min}(t) = \min_j \operatorname{clip}\left(1 - \frac{[a_{ij}(t) - A_j]_+}{B_j - A_j}, 0, 1\right), \qquad t \geq 20.$$

Here $[u]_+ = \max(0, u)$, A_j is a statistical initial tolerance and $B_j > A_j$ is an empirical terminal reference learned from fitting engines. Neither is a certified operational limit. For fixed references and channels, the margin reaches zero if and only if $a_{ij}(t) \geq B_j$; increasing A_j alone cannot correct terminal anchoring. The HI is unavailable before its initial reference, and P is not used to impose its shape.

The terminal-reconstruction comparator uses patterns $\mathcal{V}_k$ from the final observed cycles of fitting engines. Its reconstruction and residual are

$$\mathcal{K}_h(\mathbf{z}, \mathbf{v}) = \exp\left[-\frac{1}{2h^2}\sum_j \frac{(z_j - v_j)^2}{\nu_j^2}\right],$$

$$\widehat{\mathbf{z}}_F(\mathbf{z}) = \frac{\sum_{\mathbf{v}\in\mathcal{V}_k} \mathcal{K}_h(\mathbf{z}, \mathbf{v})\mathbf{v}}{\sum_{\mathbf{v}\in\mathcal{V}_k} \mathcal{K}_h(\mathbf{z}, \mathbf{v})}, \qquad D_F(\mathbf{z}) = \|\mathbf{z} - \widehat{\mathbf{z}}_F(\mathbf{z})\|_2,$$

$$\widehat{H}_i^{\mathrm{AAKR}}(t) = \operatorname{clip}\left(\frac{D_F(\mathbf{z}_i(t)) - c_F}{c_H - c_F}, 0, 1\right), \qquad c_H > c_F.$$

The scales ν_j describe terminal-pattern dispersion. The initial anchor c_H is the residual at the initial reference; c_F is the median fitting-engine endpoint residual obtained

while excluding that engine's patterns from reconstruction and metric estimation. Causal summaries and deterministic sensor-subset candidates replace the original study's feature-extraction and evolutionary-search components. Accordingly, the experiment compares an adaptation of terminal AAKR [1], not the complete published method.

S2 Empirical support and perturbation replays

The empirical-support diagnostic uses consecutive raw sensor increments, not HI values or moving averages. For each fitting engine, let $\Delta\mathbf{x}_i(t) = \mathbf{x}_i(t) - \mathbf{x}_i(t-1)$ and $\mathbf{m}_i = 15^{-1}\sum_{t=6}^{20}\Delta\mathbf{x}_i(t)$. Pool the residuals $\mathbf{r}_i(t) = \Delta\mathbf{x}_i(t) - \mathbf{m}_i$ over fitting engines and cycles 6–20. Their sample covariance $\mathbf{V}$ uses denominator $N-1$; retain channels $J = \{j : V_{jj} > 10^{-12}\}$, independently of HI channel selection, and set $\nu_j = \sqrt{V_{jj}}$. The standardised fitting residuals are $\mathbf{v}_i(t) = (r_{ij}(t)/\nu_j)_{j\in J}$. Let $\mathbf{R}$ be their sample covariance and $\mathbf{P}$ its inverse restricted to eigenvalues greater than 10^{-12}. With $Q_{.99}$ denoting the linearly interpolated empirical 0.99 quantile, define

$$L_M = Q_{.99}\{\mathbf{v}_i(t)^\top\mathbf{P}\mathbf{v}_i(t)\}, \qquad L_j = Q_{.99}\{|v_{ij}(t)|\}.$$

For an original or perturbed evaluation trajectory $\mathbf{x}_i^{(v)}$, the evaluated increment is $\mathbf{e}_i^{(v)}(t) = ([\Delta x_{ij}^{(v)}(t) - \bar{m}_j]/\nu_j)_{j\in J}$, where $\bar{\mathbf{m}}$ is the mean of the fitting-engine centres $\mathbf{m}_i$. Thus fitting residuals are centred per engine, whereas evaluation increments use the common fitting centre. An increment is inside if and only if

$$I_i^{(v)}(t) = \mathbf{1}\Big\{(\mathbf{e}_i^{(v)})^\top\mathbf{P}\mathbf{e}_i^{(v)} \le L_M, \quad |e_{ij}^{(v)}| \le L_j \text{ for every } j \in J\Big\} = 1.$$

The global perturbation adds a fixed channel-shift vector multiplied by $g(t) = \mathrm{clip}((t-20)/20, 0, 1)$. The local replay ends at the original decision C_0, or $T_i - 1$ if no original decision exists, and uses $g(t) = \mathbf{1}\{t > 20\}\,\mathrm{clip}((t - C_0 + 19)/19, 0, 1)$. Both leave the initial reference unchanged. For the global replay, each trajectory's prefix ends at its own decision, or $T_i - 1$ without a decision; in the local replay the fallback is C_0. A paired replay is wholly inside support only when

$$\prod_{t=6}^{C_0} I_i^{(0)}(t) \prod_{t=6}^{C_v} I_i^{(v)}(t) = 1,$$

where C_v is the perturbed prefix end. Destination cycle 6 and both ends are included; F is excluded. Date-shift summaries require both decisions to exist, whereas decision-presence summaries retain missing decisions. This whole-prefix condition differs from support at individual common sensitivity times. Separate 0.99 cut-offs do not imply 99% joint or whole-prefix coverage. Exceedance denotes statistical extrapolation relative to initial increments, not physical impossibility; inclusion does not establish physical plausibility.

Coefficient stability metrics

Stability of the coefficients is also measured between the folds of each repetition, from the stored models: the Spearman correlation of β_j, not w_j, over the channels retained in both folds, the share of channels with the same orientation, the Jaccard overlap of channels with $\beta_j > 10^{-9}$ and the overlap of the three largest β_j values. Ties in the latter are resolved by sensor number. This diagnostic differs from the contribution-rank

comparison above. Stability is examined by excluding each fitting engine in turn, refitting orientation and coefficients, and recording the shift of the first-passage landmark. Four alternative coefficient formulations were compared on development engines only and then through the full chain; none is adopted.

S3 Coverage of experimental gains and non-gains

This review covers the 32 scientific experimental series preceding the literal-flight analysis in S5, including the early pilots and the final confirmation and inspection analyses. Inventory and rehearsal outputs are technical checks; the failed first renewal run is preserved as an invalidated computation, not evidence of a maintenance benefit. The accompanying public package contains an experiment-by-experiment coverage matrix, source hashes and 1,531 directional contrast records. Some records describe different metrics of the same comparison, repeated engines or multiple exploratory contrasts; they are not independent discoveries. A descriptive interval excluding zero does not establish replication, population risk or practical relevance.

Series	Positive or local result	Non-gain and qualification
P01–P03	P01 margins removed observed endpoint events. In P02, local calibration within the same action region added 4.47–6.94 mean cycles with zero events across four regressors; selected LightGBM reached 196.71 mean cycles with zero events. P03 detector changes left individual removal dates invariant under fixed forecasts.	P01 margins discarded substantial life. Early pilots used immediate execution, different predictors and reused engines; they do not validate the later reserve contract. Detector invariance does not identify physical P.
P04	Initial tolerance reduced departure from the assumed healthy plateau: the departure fraction fell from 1.000 to 0.812 or 0.593.	The stable period is assumed. First-zero dates did not change with tolerance for fixed terminal references; terminal anchoring remained inadequate.
P05–P06	Refitted FD003 empirical and AAKR errors were 23.40 and 26.41 cycles against 32.41 for the margin control. Their paired differences had intervals below zero.	FD001 differences included zero. Direct transfer failed markedly; neither refitting nor good RUL error validates physical HI zero.
P07	Five-cycle updates reduced RMSE from 17.85 to 17.11 in FD001 and 14.84 to 14.31 in FD003.	At common observation cycles, the apparent FD003 advantage over the margin HI shrank to -1.05 cycles with interval including zero. The empirical HI recovered above 0.5 after crossing in 7 FD001 and 10 FD003 engines.

Series	Positive or local result	Non-gain and qualification
P08	Direct-sensor history reduced FD003 RF RMSE from 16.44 to 14.40 in complete engines and from 15.69 to 11.21 in 22 truncated engines with a landmark. In that truncated group, full sensors also improved on full HI history (15.14 to 11.21).	These are exploratory, state-conditional comparisons. Full HI history did not improve last-observation errors; Ridge worsened. HI plus slope alone gave 16.67/14.00 cycles versus 16.98/14.05 with full HI history. Individual ablation gains and losses remain in the ledger.
P09	Slope improved RF over current HI in all three FD003 repetitions, with paired intervals below zero. Path calibration improved nominal-90% RF-slope whole-path coverage from 60–82% to 86–90%.	ISO was not consistently beaten. Wider bands paid for increased coverage, and useful conservative lead could collapse.
P10–P11	Finite nominal-98% calibration was feasible in development; lower-anchor rules prevented observed endpoint events. Reserve-12 and nominal-95% sensitivities preserved more life in the same sample.	Calibration and event risk differ. Temporal RF did not conclusively outperform ISO at the main landmark; lower leads could collapse. Sensitivity gains do not authorise weaker protection.
P11.1–P11.2	The landmark supplied at least 95.9% of conservative ISO's development life gain. RF-slope added 0.28–1.09 mean cycles in FD001 and 0.24–0.54 in FD003 over that marker. Oracle comparison located residual loss after signal availability.	ISO's extra gain was 0–1.57 cycles. No loss to late access under the basic oracle does not make all post-landmark loss recoverable.
P12.1–P12.2	Analytic and simulated renewal rates agreed. Selected condition thresholds and waiting times reduced restoration demand relative to the marker; rule ordering usually survived the scale and renewal grid.	Gains of selected rules included endpoint events. Small ISO/RF or ISO/marker differences were often indistinguishable. Scale, LLP, starting age and restoration effectiveness govern the calendar scenario.
P13.1	Selected thresholds added 13.08–15.49 cycles in FD001 and 8.29–12.90 in FD003; waiting added 8.50–10.00 and 6.90–9.00.	Thresholds had 0–2 events per 100 and waiting 0–1; four internal choices were infeasible.
P13.2–P13.3	The computational explanation reproduced 9,600 actions. Influence diagnostics linked large shifts to orientation and coefficient learning; FD001 sensitivity was usually much smaller.	No whole decision prefix passed the initial-support restriction. RF forecasts were more sensitive to specified perturbations, but neither that contrast nor explanation fidelity establishes physical robustness.
P13.4	Internal FD003 median partition p95 shift fell from 5 to 3 cycles and the largest shift from 223 to 45.	Maxima worsened in 8/30 FD001 and 3/30 FD003 partitions. One contingency was not eligible; development stability was not externally validated.

Series	Positive or local result	Non-gain and qualification
P13.5	Conservative ISO gained 1.30–1.96 mean cycles in FD001 and 2.29–4.74 in FD003, with positive paired intervals. Landmark RMSE improved in 11/12 comparisons, nine with intervals below zero.	Later prediction times confound RMSE attribution. FD001 p95 range worsened and FD003 gained endpoint events; the joint transfer criterion failed.
P14.1	Relative landmark range did not increase with lifetime; longer-lived tails explained much absolute spread.	Absolute instability was only partly concentrated in long lives, while events were concentrated in short lives. No engine exclusion or universal age cap was justified.
P15.1–P15.4	Higher fallback ages recovered some life; conservative ages suppressed observed events when below all evaluated endpoints.	Suppression was partly by construction. Conservative caps largely erased PHM gains; extrapolating rejected lifetime distributions was unsupported. These superseded policies are not the monitoring-loss policy.
P16.1	Condition-based extension recovered 53%/68% of headroom in FD001/FD003 and retained some savings with detected monitoring losses.	Monitoring validity is assumed detectable. A loss-triggered removal occurs at or after the age gate, but a pre-existing PHM decision may remove earlier.
P17.1	Inspection reduced point-anchor events and FD004 conservative-rule events, with restoration-rate comparisons reported jointly.	Advancing removal costs life. Additional $\gamma = 0.26$ sensitivity increased mean restoration rates by 6.8–11.8% for lower-anchor ISO and 17.2–19.2% for point-anchor ISO relative to no inspection. Inspection counts and all curves are archived; none is borescope-calibrated.
P17.2	Low point-anchor event levels persisted under latent visibility and shape scenarios.	Heterogeneous visibility increased restoration penalties; latent shape had little effect. These are generator-conditioned findings.
P18.1	Normalisation restored synthetic multi-regime landmark availability and near-reference error, reproducing the true-regime result.	Regimes were separable and effects simple; this does not validate regime–degradation interactions.
P19.1–P19.2	The frozen age-limit criterion passed; conservative FD002 policies had no observed events and low incremental forecast gains. Coefficient stability was quantified in all subsets.	FD004 had policy events; some margins were infinite. Reserve-12 and nominal-95% ISO sensitivities did not alter FD004 lives or events. Coefficients remained only moderately stable.
P20.1	Three-arm 20-year restoration-demand and expenditure comparisons favour PHM over the fixed age in the declared central scenario and almost all grid contrasts.	Inspection increases restoration expenditure relative to the same PHM rule. Results depend on scale, renewal and cost assumptions; the failed initial simulation is excluded.

The public files retain every directional contrast rather than only the favourable intervals. Gains in a prediction metric, calibration target or development stability are distinguished from gains in realised removal performance; none selects a new policy after confirmation.

S4 Earlier scaled renewal scenarios and restoration expenditure

This section preserves the earlier scale-bridge analyses. They are sensitivity scenarios, not the literal-flight scenario now used in the main text. The sources concern engines of 19,500–33,000 lb thrust, whereas the benchmark engine class is 90,000 lb; the reference interval therefore does not physically calibrate the benchmark engine.

The developmental scale analysis used a partition-specific age control, whereas the final three-arm comparison uses the single 127-cycle benchmark gate. They are different references. With the central scale $\xi = 40.8$ in FD001 and 34.1 in FD003, 1,600 operational flight cycles per year and an LLP counter of 20,000 cycles, the earlier age control needed 6.64/7.25 visits per position over 20 years and lower-anchor ISO 5.51/5.43: reductions of 17.0%/25.1%. About 1.6 LLP visits were common to all rules. Without LLP, reductions stayed at 21.3–22.3% and 29.0–29.8% across the scale grid. ISO gains over the landmark were small; 36% of FD003 cells were indistinguishable. A literal one-to-one interpretation in the earlier, partition-age experiment gave approximately 150–240 restorations per position; that is not the reported central scenario or a new three-arm simulation.

An operational cycle is counted as engine start, flight, landing and shutdown. C-MAPSS records a snapshot per simulated flight; $\xi = L_{\mathrm{ref}}/\bar{T}$ rescales the degradation lifetime to a declared operational service-interval scenario. The benchmark gate therefore becomes 127ξ operational cycles, not 127 operational cycles. The public `cycle_scale.csv` gives both quantities by subset. There is no simulated intra-flight start or shutdown process.

All three arms share the same utilisation, horizon, initial-age condition, LLP counter and restoration assumptions. Let N_{20}^{π} count restorations after degradation removals or forced LLP removals over 20 years. At the common illustrative cost c_R, expected restoration expenditure is $c_R\mathbb{E}[N_{20}^{\pi}]$. The central c_R is US$6 million; US$5/7 million multiply every central cost by 5/6 or 7/6. The public cost table uses the unrounded archived simulation means. Costs have no discount or inflation; inspections, monitoring and downtime have no incremental charge, and endpoint events have no added consequence cost. Thus this is a comparison of restoration expenditure under a fixed-cost model, not total maintenance cost or company profit.

Inspection is not assumed costless in practice. Its zero incremental cost is the experimental contract. If engineering data later justify a nonzero inspection charge or different restoration scope after different removals, the total-cost comparison must be extended. No such new scenario or optimal policy is inferred here.

The earlier four-subset simulation used 5,760 cells over scale, utilisation, LLP, initial age and restoration effectiveness, with 2,000 replicates per cell. The first run failed its predeclared stationary check ($\max|z| = 6.15$) because the initial life distribution misweighted stochastic outcomes. That run was preserved, the weighting corrected and a test added; the corrected run passed ($\max|z| = 3.90$, long-horizon error at most 0.24%, identical LLP counts across arms). An independent scalar implementation reproduced

192 sampled cells exactly. Across the full grid, conservative ISO versus age retained the expected sign in 11,518/11,520 comparisons, and the hybrid versus the same PHM rule in 11,519/11,520; the remaining comparisons were indistinguishable. These are scenario contrasts, not independent replications.

At 1,600 flights/year, $L_{\text{ref}} = 8,425$, LLP 20,000, mature start, full restoration and lead one, the scaled three-arm costs were as follows (US$ million per position over 20 years). They use the archived unrounded means and remain distinct from the new literal-flight table.

Subset	Age 127	PHM	PHM + inspection
FD001	40.56	33.01	33.94
FD002	40.66	32.91	33.90
FD003	48.90	32.62	33.62
FD004	48.75	31.46	33.25

These totals correspond to 6.76/6.78/8.15/8.13 age visits, 5.50/5.48/5.44/5.24 conservative ISO visits, and 5.66/5.65/5.60/5.54 hybrid visits in FD001–FD004. Conservative ISO has 0/0/0/0.0235 mean endpoint events and its hybrid 0/0/0/0.0003. Point-anchor ISO has 0.3795/0.2228/0.4930/0.6160, and its hybrid 0.0142/0.0008/0/0.0008. The archived review package retains the remaining controls and exact values. Independent renewal, scale and inspection capability remain assumptions.

S5 Literal-flight renewal scenario

Following the author's explicit scenario choice, one C-MAPSS cycle is assumed to represent one flight per engine ($\xi = 1$). A twin-engine aircraft accrues one cycle at each position per flight. At 1 600 flights per year, each position accumulates 32 000 cycles in 20 years; expected restoration expenditure for two equivalent positions is twice the single-position total, without implying independent engines or an aircraft-level risk estimate. The mapping is a scenario assumption, not physical calibration of the degradation rate.

P20.2 uses only the archived complete-engine decisions: 100/180/100/169 engines in FD001/FD002/FD003/FD004 per repetition. Three repetitions reuse engines. All five PHM controls, their inspection variants and the age rule are retained for execution leads one and two. The central comparison keeps always-valid monitoring, full restoration, a mature initial fleet, an independent LLP counter of 20 000 flights and the original inspection generator (entry 127, interval five, $W = 60$, $\gamma = 1$, $p_P = 0$, $p_F = 1$). Cost is US$6 million for every restoration, including LLP, with zero incremental inspection charge and no other costs. No models, margins or removal rules were refitted or selected.

The protocol was written before this execution and after confirmation. For each of 12 subset/repetition cells, 2 000 Monte Carlo replicates share engine/outcome streams across all 22 arms. Initial lives are sampled with length weighting and residual age uniformly along the chosen life; LLP initial age is uniform on its counter. Each subsequent life draws an engine and then an outcome. The eight numerical core functions match the corrected earlier simulator. All 264 arm/cell combinations passed the stationary no-LLP check (maximum $|z| = 2.3035$, threshold 4.5); LLP visits were identical between arms in each replicate. Independent scalar calculations checked 121 406 outcome atoms and reproduced all five counts in 12 replicates per arm/cell (3 168 comparisons). Deterministic age removal was checked at both new and mature starts without LLP.

Table 2 and Table 3 give the literal-flight totals. Conservative ISO uses 149.5–191.1 restorations per position versus about 252.8 for age removal, and the same rule with inspection 159.4–197.1. The conditional mean FD004 event count falls from 0.8855 to 0.0188 per position with inspection. Point-anchor ISO has 10.4–24.8 events without inspection and 0.0018–0.6353 with it. No event outcome is removed from the restoration count. The conditional Monte Carlo standard error of any displayed mean visit count is below 0.064; it excludes population and model-selection uncertainty. The large absolute costs reflect the short assumed flight lives of the benchmark under the literal mapping and must not be interpreted as operator forecasts. The direction of the central visit comparisons agrees with the earlier scaled scenario; this does not establish physical equivalence or accepted risk.

The separate public package `literal_cycles_2026-10-19` contains selected decision inputs, a portable numerical core, the runner and independent verifier, results, conditional Monte Carlo errors and hashes. It is a new exploratory series following the 32-series review in S3, and is not part of that earlier 1,531-record ledger.

Table 2: Restoration visits and endpoint events per position over 20 years.

	Visits per position				Endpoint events per position			
Arm	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
Age limit 127	252.78	252.79	252.77	252.79	0.0000	0.0000	0.0000	0.0000
Landmark control	192.19	190.04	155.08	149.50	0.0000	0.0000	0.0000	0.8855
ISO, lower 98%	191.14	189.93	154.70	149.50	0.0000	0.0000	0.0000	0.8855
ISO, lower 98% + inspection	197.06	195.97	159.87	159.45	0.0000	0.0000	0.0000	0.0188
ISO, point anchor	167.47	170.30	138.28	141.60	18.8187	10.3587	19.1698	24.8033
ISO, point anchor + inspection	188.25	188.65	153.93	158.10	0.6353	0.0633	0.0018	0.0662

Literal-flight scenario ($\xi = 1$), 32 000 flights per position, execution lead 1, mean over three repetitions that reuse engines.
Endpoint events are simulated mean counts of restorations following an arrival without decision, an exact arrival or a strict late schedule. Zero denotes no event in the simulated replicates, not zero population risk. Event counts are descriptive simulation means; inspection capability and the one-cycle-per-flight mapping remain hypothetical.

Table 3: Illustrative restoration expenditure per position over 20 years (US$ million).

Subset	Age 127	PHM	PHM + inspection
FD001	1516.70	1146.82	1182.36
FD002	1516.77	1139.59	1175.85
FD003	1516.61	928.22	959.22
FD004	1516.74	897.02	956.68

Same literal-flight scenario and ISO lower-98% rule as Table 2; US$6 million per restoration, including forced LLP visits, with zero incremental inspection cost. Values use unrounded simulation means; the largest Monte Carlo standard error for these three arms is US$0.38 million per position. Costs of US$5 or 7 million scale every entry linearly and preserve relative savings.

S6 Break-even thresholds under alternative inspection assumptions

Table 6 gives the effect of a positive inspection cost; the next two tables vary the damage window and curve with free inspection.

Break-even additional cost β^* of an end-of-life event, in overhaul units, for execution lead one (ranges over three repetitions that reuse engines). $<$ means that A is cheaper only below the threshold; † marks thresholds whose avoided-event interval includes zero in at least one repetition. Latent rows use an engine-level random effect on the damage window; S1 enters inspection at $T - W$ instead of at the 127-cycle gate.

Table 4: Inspection added to the ISO point-anchor rule (A) against the same rule without inspection (B).

Scenario	FD001	FD002	FD003	FD004
$W = 30$	0.23–0.38	0.30–0.39	0.16–0.22	0.14–0.26
$W = 60$ (main)	0.94–1.72	1.51–1.95	0.67–0.96	0.51–1.27
$W = 80$	1.52–2.83	2.52–3.26	1.09–1.56	0.80–2.14
$\gamma = 2$	0.54–0.96	0.83–1.05	0.38–0.55	0.30–0.70
$\gamma = 0.26$	1.44–2.67	2.38–3.10	1.02–1.46	0.74–1.98
$p_P = 0.02$	1.28–2.36	2.14–2.77	1.40–2.02	0.92–2.52
S1 entry, $W = 60$	1.04–2.07	1.86–2.42	0.72–1.03	0.55–1.40
latent window $\sigma = 0.25$	0.98–1.80	1.59–2.05	0.70–1.01	0.53–1.35
latent window $\sigma = 0.5$	1.06–1.96	1.76–2.27	0.79–1.13	0.59–1.56

Table 5: Inspection added to the ISO lower-98% rule (A) against the same rule without inspection (B); B dom. means no event is avoided.

Scenario	FD001	FD002	FD003	FD004
$W = 30$	B dom.	B dom.	B dom.	0.67–1.19† or B dom.
$W = 60$ (main)	B dom.	B dom.	B dom.	5.84–11.8† or B dom.
$W = 80$	B dom.	B dom.	B dom.	11.3–23.0† or B dom.
$\gamma = 2$	B dom.	B dom.	B dom.	2.74–5.75† or B dom.
$\gamma = 0.26$	B dom.	B dom.	B dom.	10.2–19.9† or B dom.
$p_P = 0.02$	B dom.	B dom.	B dom.	14.0–29.4† or B dom.
S1 entry, $W = 60$	B dom.	B dom.	B dom.	6.71–12.6† or B dom.
latent window $\sigma = 0.25$	B dom.	B dom.	B dom.	6.49–13.3† or B dom.
latent window $\sigma = 0.5$	B dom.	B dom.	B dom.	8.08–16.9† or B dom.

Table 6: Break-even cost β^* with a positive inspection cost $c_I = q_I c_R$ (execution lead one, main inspection scenario; ranges over three repetitions that reuse engines).

A vs B	q_I	FD001	FD002	FD003	FD004
Inspection on ISO point	0.005	1.33–2.48	2.24–2.90	1.23–1.76	0.86–2.35
	0.01	1.72–3.24	2.97–3.84	1.79–2.56	1.22–3.42
	0.02	2.49–4.75	4.43–5.73	2.91–4.16	1.93–5.57
Inspection on RF-slope point	0.005	1.90–2.84	3.13–3.40	2.18–3.31	1.62–3.23
	0.01	2.49–3.73	4.17–4.56	3.21–4.89	2.35–4.73
	0.02	3.67–5.50	6.25–6.90	5.27–8.06	3.79–7.74
Inspection on ISO lower-98%	0.005	B dom.	B dom.	B dom.	12.9–27.0† or B dom.
	0.01	B dom.	B dom.	B dom.	20.0–42.1† or B dom.
	0.02	B dom.	B dom.	B dom.	34.1–72.4† or B dom.
ISO point + inspection vs lower-98%	0.005	B dom.	B dom.	B dom.	12.8–25.5† or B dom.
	0.01	B dom.	B dom.	B dom.	20.3–41.0† or B dom.
	0.02	B dom.	B dom.	B dom.	35.2–71.9† or B dom.

B dom.: no $\beta \geq 0$ favours A. †: the 95% interval of the avoided events includes zero in at least one repetition. Inspection time and corrective repair remain zero. $q_I = 0$ reproduces the break-even analysis of the main text.

S7 Development results summarised in the main text

This section preserves development results that the main text only summarises: the empirical HI candidates and their transfer, post-landmark updates and temporal features, selected HI thresholds and waiting times, the synthetic multi-regime normalisation test and a pointer to the early immediate-action pilots.

Empirical HI candidates and transfer

Minimum-margin HIs exhibited early zero crossings, missing crossings and subsequent recoveries; adding initial tolerance did not resolve terminal anchoring (Table 7); the labels P03–P06 identify the exploratory pilot rounds documented in the project repository. These failures motivated the empirical combination and terminal-reconstruction comparator, without identifying an empirical HI zero with physical F.

Table 7: Development diagnostics for empirical HI candidates.

Pilot/subset	HI	Crossings	Missing	Median lead	Recoveries
P03/FD001	Minimum, $q = 0.50$	94	6	42.5	30
P03/FD001	Minimum, $q = 0.90$	41	59	5.0	5
P04/FD001	Tolerant minimum	81	19	21.0	15
P05/FD001	Empirical	71	29	14.0	0
P05/FD001	Terminal AAKR	92	8	18.5	43
P06/FD003	Empirical, refitted	46	54	5.0	0
P06/FD003	Terminal AAKR, refitted	82	18	18.0	52

Complete outer-evaluation trajectories, with 100 engines per row. Median lead is endpoint minus first HI zero, conditional on a crossing, in cycles. Changes between rounds include selection rules and are not isolated causal effects of representation.

P05 compared the candidates through the same separately calibrated HI-to-RUL map. Official FD001 RMSE was 28.85 cycles for the margin control, 27.66 for the empirical combination and 27.57 for adapted terminal AAKR (Table 8). Descriptive paired bootstrap intervals for differences from the control included zero: $[-4.05, 1.65]$ cycles for the empirical combination and $[-3.90, 1.43]$ for AAKR. These results establish a comparison procedure, not a conclusive winner or accurate reconstruction of physical condition.

Table 8: Official-test RUL prediction errors (cycles).

Evaluation	Margin control	Empirical	Terminal AAKR
FD001, fitted on FD001	28.85	27.66	27.57
FD003, FD001 transfer	38.58	79.56	80.25
FD003, fitted on FD003	32.41	23.40	26.41

Root mean square error at the last observed cycle of each official test engine. Each row evaluates 100 engines. FD003 transfer and refitting use the same target engines; FD001 and FD003 are different populations.
AAKR denotes auto-associative kernel regression, implemented here as a terminal-core adaptation.

Refitting on FD003 reduced empirical-combination RMSE from 79.56 to 23.40 cycles and terminal-AAKR RMSE from 80.25 to 26.41, with paired differences $[-68.56, -43.85]$ and $[-66.92, -41.52]$ cycles. The selected empirical target changed from $(\rho, \eta, \lambda) =$

$(100, 1, 0.1)$ to $(60, 2, 0)$. Refitting changed both HI and converter, while median life changed from 199 to 220.5 cycles; their effects cannot be separated.

Against the refitted margin control, the empirical and AAKR RMSE differences were -9.00 cycles ($[-14.11, -3.63]$) and -6.00 ($[-10.97, -0.89]$), respectively. These descriptive paired intervals favour the alternatives in the official FD003 test, without establishing superiority beyond those implementations.

After refitting, the empirical HI reached zero in 46 complete FD003 evaluation trajectories, with median anticipation of five cycles and median terminal HI 0.0163. AAKR crossed more often but recovered in 52 engines. Neither validates physical F.

Post-landmark updates and temporal features

Updating five cycles after the landmark reduced empirical-HI forecast RMSE from 17.85 to 17.11 cycles in FD001 and from 14.84 to 14.31 in FD003. However, comparisons at family-specific landmarks mix representation and timing: synchronising the margin and empirical HIs reduced their FD003 RMSE difference to -1.05 cycles, with interval $[-2.96, 0.69]$. With RF in FD003, temporal sensor statistics reduced RMSE from 16.44 to 14.40 in 100 complete engines (-2.04; $[-3.63, -0.42]$) and from 15.69 to 11.21 in the 22 truncated engines with a landmark (-4.48; $[-8.23, -1.98]$). These gains were local to the evaluated states: HI-history gains did not persist at the last official observation, Ridge did not improve, and slope alone matched or slightly improved on the full HI history. All contrasts remain exploratory.

Selected HI thresholds and waiting times

A threshold or a waiting time on the HI, selected from development engines alone, added 13.08–15.49 cycles (FD001) and 8.29–12.90 (FD003) to the marker for the threshold, and 8.50–10.00 and 6.90–9.00 for waiting, across repetitions and both execution leads. Threshold rules had 0–2 endpoint events per 100 engines in each subset, and waiting-time rules 0–1; four of the 240 internal selections were infeasible. These controls recover more life than immediate landmark removal, but neither their gains nor those of the point-anchor forecast establish a preferable policy under a common validated risk constraint.

Synthetic multi-regime normalisation test

With one regime the normalisation reproduced the original FD001 HI and forecasts in all ten folds, with a maximum forecast difference below 10^{-7} cycles. In a synthetic test with six regimes injected into FD001, the raw pipeline lost the landmark in 27% of the engines and increased the landmark RMSE by 66% (Table 9). With the normalisation, performance returned to within 1.4% of the original. K-means recovered the injected regimes without error. This is a necessary and not a sufficient condition, because the regimes were well separated and the effects were additive and multiplicative per sensor. Interaction between regime and degradation is not represented.

Early immediate-action pilots

The immediate-action pilots P01 and P02, including the local and global margin calibrations and the internally selected LightGBM regressor [2] cited in the main text, are

Table 9: Synthetic multi-regime normalisation test on FD001.

Variant	Landmark reached	Landmark RMSE	Spearman, HI vs RUL
Original FD001	100%	16.14	0.992
Synthetic, raw	73%	26.86	0.755
Synthetic, K-means normalised	100%	16.36	0.992
Synthetic, true regimes	100%	16.36	0.992

Ten folds of 10 evaluation engines (100 engines). The landmark is the first HI passage at 0.5.

specified and reported in full in the companion document `early_pilots.pdf` of this repository; Section S3 summarises their gains and limitations.

S8 Further method and result details

The following paragraphs are retained in full from earlier versions of the main text, which now summarises them.

All 9 600 original action prefixes contained an increment outside initial empirical support. Consequently, no paired prefix met the whole-prefix inclusion condition among the 1 084 160 nonzero global replays or the same number of local replays. These are engine–partition–policy–perturbation comparisons, not independent engines. Both decision dates existed in 1 059 726 global and 840 064 local comparisons; none qualified for a support-restricted date-shift estimate. This establishes statistical extrapolation relative to initial increments, not physical implausibility, and does not validate physical robustness.

Excluding single fitting engines flipped five sensor orientations in 13 of 25 cases in one repetition and moved the landmark by 204–213 cycles. Coefficient refitting alone shifted it by up to 179; optimality checks did not indicate numerical failure. Internal stabilisation reduced the median partition p95 shift from five to three cycles and the largest shift from 223 to 45 in FD003, but worsened maxima in 8/30 FD001 and 3/30 FD003 partitions. In the full-chain evaluation, selected variants gained 1.30–1.96 mean cycles in FD001 and 2.29–4.74 in FD003 for conservative ISO, with positive paired intervals. Landmark RMSE improved in 11 of 12 contrasts, partly with later prediction times. Between-repetition p95 landmark range changed from 9.15 to 12.05 cycles in FD001 and from 27.45 to 21.60 in FD003, and FD003 gained endpoint events. These mixed results failed the pre-specified transfer criterion; the working HI was retained.

A post-confirmation descriptive check of the archived ISO decisions separated finite and infinite margins without refitting. Across repetitions and both execution leads, the finite-margin rule acted at its first usable opportunity in 55–86% of FD001 evaluations, 72–100% in FD003 and 90.6–100% in FD002. The corresponding incremental mean life over the marker was 0.21–1.57, 0–0.83 and 0–0.29 cycles. In FD004, first-opportunity actions accounted for 99.3–100% of finite-margin evaluations; all 17 infinite-margin evaluations per execution lead acted immediately, and both groups had zero incremental life. The denominator includes engines without a decision. Thus, the limited gain also occurs with finite margins; infinite margins enforce landmark removal for part of FD004 but do not alone explain the finding. This analysis describes the tested entry rule, reserve and calibration rather than the value of forecasts under other policies.

When the visibility of the damage varies between engines, so that inspections of the same engine are dependent, the cost of the arm rises. With a window varying by $\sigma = 0.25$, the rate increase of the lower-bound rule was 3.9%, 3.9%, 4.1% and 7.5% in FD001 to

FD004 (3.1–6.7% with independent inspections), and with $\sigma = 0.5$ it was 5.3%, 5.5%, 5.9% and 9.3%; for the point-anchor rule it rose from 10.8–12.5% to 11.4–14.1%. A latent shape had little effect. The expected events of the point-anchor rule with inspection stayed below 0.4 per 100 engines in every variant. The qualitative conclusion is unchanged, although differences in damage visibility increase the restoration-rate penalty.

The five-cycle interval is a frequent-inspection scenario in simulated time, not an interval derived from an approved engine maintenance programme. European Union Aviation Safety Agency (EASA) airworthiness directive (AD) 2019-0249R1 requires inspections of specified Trent 1000 intermediate-pressure compressor parts within five flight cycles after defined asymmetric-power or depressurisation events [3]. This is an event-triggered compliance deadline, not a repetitive five-cycle inspection interval, and does not calibrate the inspection model used here. By contrast, EASA AD 2025-0206R1 imposes repetitive borescope inspections of specified Trent 1000 high-pressure turbine blades at no more than 50 flight cycles, reduced to ten after defined crack findings [4]. The five-flight cadence therefore lies below the documented repetitive range and represents a favourable inspection-capability scenario rather than a representative interval; its admissibility rests on the declared damage window, since five cycles plus the reserve and execution lead remain below the smallest window tested.

The one-cycle-per-flight assumption used for the illustrative 20-year totals does not physically calibrate the inspection cadence. The interval remains fixed at five benchmark cycles in the frozen comparison, interpreted as five flights in the literal scenario. Its influence has not been assessed by interval sensitivity analysis, which, with representative inspection costs and downtime, would be needed before operational interpretation. The reported events, utilised life, inspection counts and restoration demand are conditional on the assumed cadence and rejection model and do not establish an operational inspection schedule.

S9 Details moved from the main text for length

The paragraphs and tables below are retained in full from the version of the main text that preceded the length reduction for submission.

Each HI family uses an increasing isotonic converter [5] fitted on separate engines, $\widehat{r}_i(t) = g_{c,k}(\widehat{H}_i^c(t))$, with $c \in \{\min, \mathrm{emp}, \mathrm{AAKR}\}$. Development selection minimises RUL error at 50%, 70% and 90% of each complete trajectory. The three retrospective locations span horizons from an earlier prognostic opportunity to near the recorded endpoint, without tying the choice to one instant. Sampling is confined to complete development trajectories: the fractions are not physical degradation boundaries, and operational prediction and removal do not use the eventual lifetime. The development folds use 40 fitting, 20 converter, 20 selection and 20 evaluation engines. Final official-test comparisons use 60/20/20 training engines, with choices fixed before test scoring. Test labels score only the last observed cycle, without reconstructing future sensors.

FD003 has one operating condition and two degradation modes, involving the high-pressure compressor and fan. Direct transfer preserves FD001 parameters and converters; refitting learns them from FD003 training engines. Lifetime distributions also differ, so this exploratory contrast does not isolate the effect of the second mechanism.

$\alpha = 0.5$ was set at the exploratory midpoint of the bounded empirical scale, between HI = 1 and HI = 0, so that forecasting could start from a reproducible intermediate condition. P, which is capability-dependent and is not identified in C-MAPSS, need not

be estimated for that choice. The value is the numerical midpoint of the index, not the temporal midpoint of the P–F interval, half the remaining useful life, a reliability of 50%, or a physically validated degradation threshold. Even the conventional training target satisfies $H_i^* = 0.5$ at RUL $= \rho\, 2^{-1/\eta}$, which depends on the chosen scale. Adequacy is read from landmark availability, remaining lead to the recorded endpoint, error in predicted F, later prediction updates, sensitivity to the HI formulation and the removal outcomes that follow.

The sum is rectified before the HI is clipped; individual contributions remain signed. For a decomposition of the bounded loss, the radial allocation is $a_{ij} = c_{ij}\,\mathrm{clip}(S_i, 0, 1)/S_i$ when $S_i > 0$ and zero otherwise, giving $\sum_j a_{ij} = 1 - \widehat{H}_i$. Explanations are assessed for their reproduction of the HI, converter and action chain.

Coefficient stability uses foldwise ranks, orientation agreement, active-channel overlap and the three largest coefficients; leave-one-engine-out refits assess landmark shifts. Supplementary Material S2 gives the exact metrics. Alternatives were selected in development before full-chain evaluation.

The empirical-support diagnostic compares consecutive raw sensor increments with the initial fitting-engine increment distribution, using simultaneous Mahalanobis [6] and channelwise limits. A paired perturbation is classified as wholly inside only if every increment in both original and perturbed decision prefixes passes both limits. Supplementary Material S2 gives the centring, covariance, cut-offs and replay definitions. These are empirical extrapolation diagnostics; inclusion does not establish physical plausibility, and separate 0.99 cut-offs do not imply 99% whole-prefix coverage.

A 20-year horizon is used only to illustrate repeated exposure. With one cycle per flight ($\xi = 1$, not calibrated), 1 600 flights per year and an independent LLP counter of 20 000 flights [7], each position accumulates 32 000 cycles; the simulation, its validation and the earlier scaled scenarios, which use $\xi = L_{\mathrm{ref}}/\bar{T}$ with a reference interval of 8 425 flight cycles from maintenance-cost analyses [8, 9, 10], are in Supplementary Material S4 and S5.

For illustration only, at one cycle per flight and 1 600 flights per year, removal at 127 flights implies about 253 restorations per position in 20 years against 149–191 for the conservative ISO rule, and 159–197 with inspection added (Supplementary Material S5). These totals follow arithmetically from the rates, depend on the cycle mapping and are implausibly high for real engines; only the comparisons between arms are interpreted.

In the earlier uncertainty experiment, adding slope improved RF over current HI in each FD003 repetition, with descriptive paired intervals excluding zero, but did not consistently outperform ISO. For nominal 90% bilateral bands, calibrating the maximum error along the trajectory increased FD003 RF-slope whole-path coverage from 60–82% to 86–90%, at the cost of wider bands, with no evidence on removal risk.

A complementary diagnostic of the stored models includes all four subsets; the FD002 and FD004 models come from the confirmation reported in the main text. Between folds of the same repetition, the median Spearman correlation of β_j was 0.28–0.35 (FD001), 0.51–0.67 (FD002), 0.47–0.51 (FD003) and 0.39–0.50 (FD004). The orientation was the same in all retained channels for FD001 and FD002, in 94% for FD004 and in 100% for FD003 except in the first repetition, where it fell to 67%. The overlap of channels with $\beta_j > 10^{-9}$ was 0.79–0.86, 0.63–0.73, 0.44–0.50 and 0.40–0.46, and the three largest β_j values agreed in a third to two thirds. The coefficients are therefore only moderately stable, the less so in the subsets with two fault modes, and an explanation by individual sensor coefficients should be read with that variability. Repetitions reuse the same engines,

so these figures are not independent.

The instability of the landmark was only partly concentrated in long-lived engines, and not enough to justify excluding any engine. The Spearman correlation between life and the between-repetition range of the landmark was 0.171 (FD001; interval including zero) and 0.283 (FD003); relative range did not grow with life, and endpoint events, counted descriptively, were more frequent in short lives. No engine was excluded. A single age limit of $A_{\mathrm{L}} = 127$ cycles was set. None of the 200 examined engines has $T_i \leq 127$, so the upper confidence limits for $\Pr(T \leq A_{\mathrm{L}})$ are 0.80%, 1.14% and 1.49% at confidence 80%, 90% and 95%. For FD001 alone the 95% limit is 2.95%. These limits are in-sample.

The landmark of an engine varied by a median of 6 (FD002) and 9 (FD004) cycles across repetitions, with maxima of 31 and 40, without a range comparable with that of engine 55 of FD003.

Fixed reserve-12 and nominal-95% sensitivities increased utilised life in development without observed endpoint events (the main text). In FD002 they added only small gains; in FD004 conservative ISO lives and events were unchanged. These sensitivities were not selected as replacements for the main rules; Supplementary Material S3 and the public contrast tables retain them.

With the extension over the monitoring-loss grid, the lower-bound ISO rule keeps a saving against the 127-cycle limit of 24.5–25.3% (FD002) and 40.9–41.2% (FD004) with always-valid monitoring, 21.4–22.1% and 33.8–34.0% at $u = 0.005$, and 15.1–15.4% and 21.3–21.5% at $u = 0.02$. The pattern reproduces the development subsets, and the larger saving in FD004 reflects its longer lives.

Three candidate tasks are evaluated: removal at 127 cycles; sensor-only removal from the landmark, forecast anchor, reserve and execution lead, with the monitoring-loss response in the main text; and the hybrid addition of five-flight inspections from flight 127 under the literal mapping, where rejection can only advance removal. Each specifies entry, trigger, execution lead and conduct on monitoring loss.

Within the assessment of alternative tasks framed by IP-180, the present evidence covers the benchmark mechanisms and their declared scenarios. IP-180 limits AHM use to tasks that are not safety-related and do not cover certification maintenance requirements, and excludes AHM-supported evolution of Maintenance Review Board Report (MRBR) task intervals [11]; Issue Paper 197 (IP-197) assigns certification to on-aircraft features and validation of ground processing to the operator's authority [12]. Because the C-MAPSS endpoint represents exhaustion of operability margins, the failure-effect category of the removal task, and hence its eligibility under IP-180, remains undetermined. Physical failure limits, representative inspection capability, operational monitoring validity and engineering evidence for task applicability also remain to be established. The comparison therefore provides candidate task formulations and identifies the evidence needed for further assessment; it does not complete an IP-180 applicability or approval process.

Separate calibration and reserve contrasts

An exploratory paired analysis of archived ISO decisions separates nominal calibration (98% versus 95%) and reserve (15 versus 12), with the same engines, folds, landmarks and execution leads. Table 10 spans three repetitions and both leads; the ranges are estimates, not confidence intervals. All four combinations had zero strict late schedules, exact arrivals and absent decisions in the evaluated FD001 and FD003 engines. Both factors increased life, with a positive interaction; this does not certify population-risk equivalence. The joint

change added 2.4–7.0 mean cycles over the HI marker across both leads. The portable analysis and per-repetition contrasts are in `experiments/sa3_20261020/`.

Table 10: Separate effects of calibration and reserve in development: A minus B.

Contrast	Mean-life change		Restoration-rate change	
	FD001	FD003	FD001	FD003
95%–98%, reserve 15	0.73–1.81	0.51–1.55	−0.639 to −0.254	−0.350 to −0.119
95%–98%, reserve 12	1.90–3.25	1.28–2.58	−1.093 to −0.648	−0.577 to −0.296
Reserve 12–15, 98%	0.50–2.65	0.49–1.82	−0.931 to −0.174	−0.412 to −0.111
Reserve 12–15, 95%	2.23–4.04	1.52–2.27	−1.377 to −0.758	−0.503 to −0.338
Joint: 95%/12–98%/15	3.39–5.39	2.39–3.30	−1.836 to −1.160	−0.735 to −0.535
Life interaction	0.66–1.92	0.33–1.43		

Life is measured in simulated cycles and restoration rate per 10 000 cycles. The interaction is the calibration effect at reserve 12 minus that at reserve 15. Ranges must not be added across different repetitions or leads.

Table 11: Extension beyond 127 cycles under valid monitoring.

Rule	FD001		FD003	
	Extension	Saving	Extension	Saving
ISO, lower anchor	41.2	24.5%	81.2	39.0%
HI-marker control	40.3	24.1%	80.7	38.8%
Selected threshold (exploratory contrast)	55.1	30.2%	91.8	41.9%
ISO, point anchor	65.1	33.9%	106.0	45.5%
ISO lower, $u = 0.005$		21.6%		31.6%
ISO lower, $u = 0.02$		15.5%		19.9%

Execution lead 1; mean over three repetitions of the 100 evaluation engines. Saving is the relative reduction in restorations per simulated cycle against removal at $A_{\mathrm{L}} = 127$ cycles.
The selected-threshold row is an exploratory contrast outside the main rule set. The last two rows give the ISO lower-anchor saving when monitoring is invalid at each cycle with probability u.

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Table 12: Borescope inspection outcomes in the headline scenario.

Subset	Rule	Rate change (%)	Life ceded	Events alone	Events with inspection	Rejection share
FD001	ISO, lower 98%	3.1 (2.3–4.0)	5.1	0.00	0.00	0.45
FD001	ISO, point anchor	12.5 (10.6–14.5)	21.3	11.33	0.34	0.92
FD002	ISO, lower 98%	3.2 (2.5–4.0)	5.2	0.00	0.00	0.45
FD002	ISO, point anchor	10.8 (9.7–11.9)	18.4	6.11	0.03	0.89
FD003	ISO, lower 98%	3.4 (2.6–4.3)	6.8	0.00	0.00	0.53
FD003	ISO, point anchor	11.4 (10.0–12.9)	23.8	14.00	0.00	0.94
FD004	ISO, lower 98%	6.7 (5.7–7.8)	13.5	0.59	0.01	0.76
FD004	ISO, point anchor	11.7 (9.5–14.0)	23.9	17.75	0.04	0.91

Entry at the 127-cycle gate every five cycles, $W = 60$, $\gamma = 1$, $p_P = 0$, $p_F = 1$, always-valid monitoring, execution lead 1.
Values are means over three repetitions that reuse the same engines; the rate change is the relative increase in restorations per simulated cycle against the same rule without inspection, with the range of the 95% bootstrap intervals across repetitions.
Events are expected strict late schedules, exact arrivals and arrivals without decision per 100 engines.
Rejection share is the mean probability that the decision comes from a rejection.

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